

Venkata Bhanu Teja Pallakonda

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SUMMARY

AI researcher and engineer specializing in **LLM post-training**, **agentic AI systems**, and **edge AI** for critical and remote infrastructure. Inventor on **3 granted U.S. patents** commercialized in a shipping product, with peer-reviewed research in edge LLMs, AI security, and geospatial ML. Currently lead model post-training and agent harness engineering at Armada.

EXPERIENCE

Lead AI Engineer

Armada AI

Jul. 2023 – Present

Seattle, Washington

First AI hire; **lead model post-training** (SFT/RL) and **agent harness engineering**, owning the AI stack for Armada's edge platform – from NVIDIA H100/A100 GPUs to CPU-only nodes. Inventor on **3 granted U.S. patents** (assignee: Armada Systems), all in production.

Conversational AI Assistant. Built the core **AI Assistant** from scratch on open-weight LLMs (**Llama 2/3**) with an **LLM-as-router** design; deployed end-to-end on **Azure Kubernetes Service** with **vLLM** and **quantized** models (llama-cpp) for on-device edge inference.

Multi-agent systems. Designed a production **multi-agent system** (supervisor-sub-agent) and modular **inference routing** across API and self-hosted endpoints; hardened against **prompt-injection** with agent observability & evaluation (Langfuse, Grafana, CI). Shipped **OPUS** for **PTZ camera control** – +35% over advanced prompting and +20% task accuracy vs. closed-source via SFT on synthetic data (IEEE ICCAR 2025).

Skill-agent harness & coding agents. Engineered a **skill-agent harness** – LangChain **deep agents**, **code sandboxing**, and Armada-specific **asset skills** – and re-architected the AI Assistant as a composable **skill + coding agent**. Led **AI-safety** research on backdoors in tool-using LLMs (RL fine-tuning, distributed multi-GPU; *Sleeper Cell* – arXiv:2603.03371).

SELECTED PUBLICATIONS

Sleeper Cell: Injecting Latent Malice Temporal Backdoors into Tool-Using LLMs 2026

B. Pallakonda, M. Hindsbo, S. Ehsani, P. Mishra | arXiv:2603.03371 | To be submitted, NeurIPS 2026

Camera Control at the Edge with Language Models for Scene Understanding 2025

A. Buynitsky, S. Ehsani, B. Pallakonda, P. Mishra | IEEE ICCAR 2025 | DOI

Mapping the Storm: Geospatial Impacts of Severe Weather on LEO Network Performance 2025

S. Ehsani, B. Pallakonda, P. Mishra | ACM SIGSPATIAL 2025 | DOI

PATENTS

All entitled “*Edge Computing Units for Operating Conversational Tools at Local Sites*” (assignee: Armada Systems Inc.).

- **US 11,960,515** – granted Apr. 2024 (lead patent of the family)
- **US 12,001,463** – granted Jun. 2024
- **US 12,093,294** – granted Sep. 2024
- **US App. 18/810,025** – pending

EDUCATION

M.S. in Computer Science, Texas A&M University

Aug 2021 – Jun 2023

B.Tech. in Electrical Engineering, Indian Institute of Technology (IIT) Tirupati

Aug 2015 – May 2019

TECHNICAL SKILLS

Research	LLM Post-Training (SFT, RLHF/RL), Agentic Systems (Multi-Agent, Skill & Coding Agents), Evaluation & Red-Teaming, RAG, CV, NLP
Programming	Python (PyTorch, TensorFlow, FastAPI, LangChain/LangGraph, Deep Agents), JavaScript, React, SQL
Systems & Infra	Agent Harnesses, Code Sandboxing, vLLM, llama-cpp, Langfuse, Azure/AKS, Docker, Kubernetes, CI/CD, Edge

EARLIER EXPERIENCE

Machine Learning Engineer, Fincare Small Finance Bank, Bangalore

Jun 2019 – Oct 2020

Built a **BERT**-based banking chatbot (intent & entity), ID-card detection, and privacy-preserving field-masking pipelines in production.

Research Assistant (NSF), Texas A&M & **ML Intern**, Productiv

2022

Pancreatic-cancer prediction via feature engineering; LayoutLMv3-based contract field-extraction.